Intro to Sensor Calibration with Arduino and Pressure Transducer

Objective: Learn how to program an Arduino to complete analog sensor calibration

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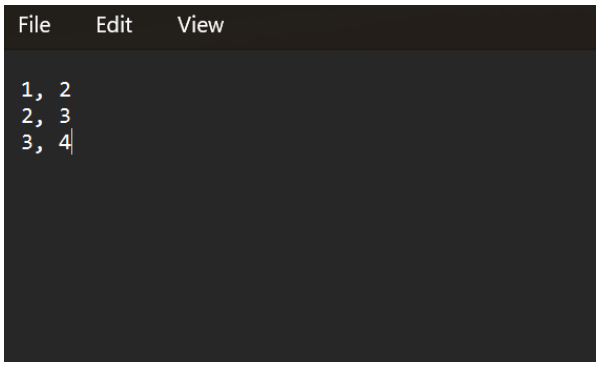
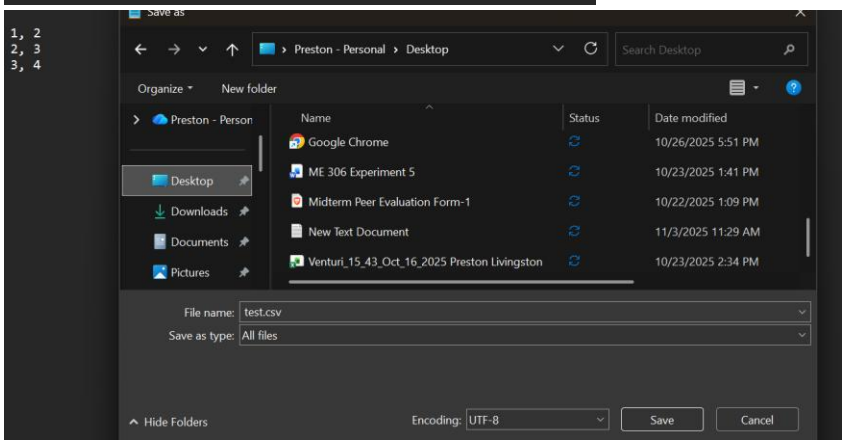
Required Items:

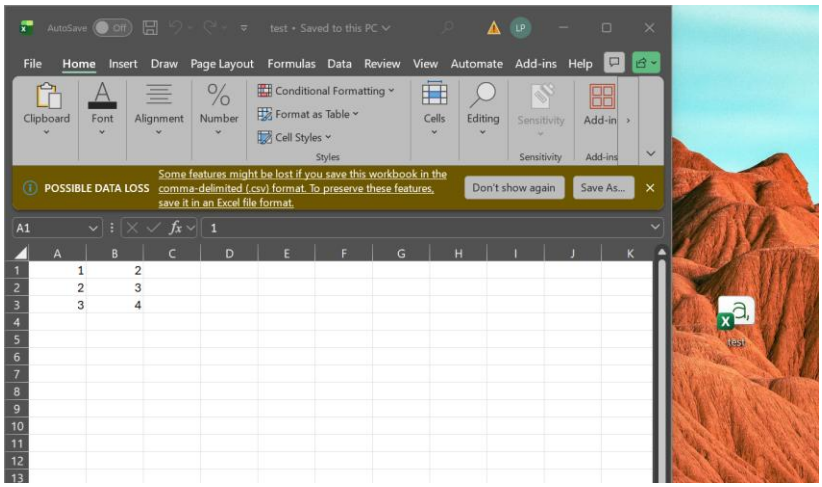
- Pressure Transducer
    - Need to find data sheet for pressure transducer as well
- Arduino Micro controller
- Laptop
    - Arduino IDE should be installed as well
- Known Pressure Source

1. Find Data Sheet for Pressure Transducer
    a. In this sheet it should tell you a few things about the sensor. For our purposes the most important things are: maximum pressure rating, output voltage range (typically 0.5-5v), and power supply (typically 5v).
    b. Note all those values down as some of them may become more important later
2. Writing the routine in Arduino IDE
    a. Arduinos and similar micro controllers typically operate off of code written in C++, if you are unfamiliar with C++ view this link: https://www.w3schools.com/cpp/
        i. C++ is an explicitly typed language, unlike Python which is dynamically typed, statements need to have explicit terminations, most of the time it will be a ";" and each and every data type must be defined, please use the link above it is a great resource for learning new languages and talks about this in depth.
        ii. A code template will also be provided on the GitHub
            1. There will be more info there
    b. There are specific functions exclusive to Arduino that will be needed for this project, please view this link: https://docs.arduino.cc/language-reference/en/functions/communication/serial/
        i. Serial Port commands are what allow your computer to communicate with the microcontroller, it is a UART connection which isn't too fast, but it is more than enough for what we will be doing

c. Please feel free to reach out to me either in person or via discord I will do my best to help during this. Also do not let this stress you out, this is purely just to learn a new skill, there are no strict deadlines with this.

3. Test your code with no pressure transducer and make sure it behaves as you expect it to.
4. Show me the code, if it's good we will move on to the calibration steps, we will be using a water pump for this, we will collect ADC(Analog Digital Conversion) counts and voltage readings.
    a. Please have your code printed data out in this format. ADC, Voltage. The reason we print data out in this format is that we can copy and paste into a txt file then save as a csv which will create an excel sheet.
        i. 

        ii. 

iii. 

b. While using the water pump keep track of how much pressure the gauge reads at each data collection point, record these "true pressure" values on a note sheet.
c. Once all data is collected and saved to a csv, place all your recorded true pressure values in the excel file as well.
d. Now make 2 plots with a linear best fit line
    i. True Pressure (y-axis) vs. Vout – Vmin (x-axis)
    ii. True Pressure(y-axis) vs. ADC
e. The equations of these plots are the calibration equations, either of them should work but it never hurts to have options

5. Congrats you have calibrated pressure transducers!!